

EMPLOYEE MANAGEMENT SYSTEM(EMS)



CSE 100: Software Development Project I

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Abstract

[The “Employee Management System (EMS)” project is a console-based software developed in C++ to manage employee-related tasks in an organization. It provides easy access to employee data and automates key processes, improving efficiency. The system offers a simple interface with two user roles: Administrator and Guest, each with different access levels. Key features of the EMS include searching and viewing employee data, saving and loading files for easy data management, and a salary grading system based on predefined criteria. It also allows storing and retrieving detailed employee information, helping HR managers and administrators track employee performance, roles, and compensation. The system provides customizable viewing options for better presentation of data. The advantages of the EMS include better organization and accuracy in managing employee data, simplified administrative tasks, and increased security through user roles. It reduces manual work and errors, improving overall productivity and ensuring that employee information is always accessible and up-to-date, aiding decision-making and efficient management.

]

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Chapter 1

INTRODUCTION

1.1. Problem Specification

Manual handling of employee information presents several challenges, such as slow processes in leave management, where approval can take weeks. Paper-based systems are prone to human error, loss, and unauthorized access, while being time-consuming. Many current systems lack employee self-service, requiring staff to go through an administrator to access their own data. Additionally, multinational companies face difficulties accessing employee information stored at headquarters, especially in remote locations. This project aims to create a secure, password-protected employee information system to store details such as personal information and salary, enabling efficient management and easy access to important data.

[1]

1.2. Objectives

The main objectives of the Employee Management System(EMS) are:-

- To develop a user-friendly, console-based application to manage employee information.
- To automate tasks like storing, retrieving, and updating employee data, reducing manual errors.
- To implement a salary grading system based on predefined criteria, improving payroll management.
- To provide different access levels (Administrator and Guest) to ensure secure and appropriate access to employee data.
- To keep employee information up-to-date and easy to access for decision-making.

1.3. Scope

- Project Goals: Automate employee data management, implement user roles, create a 12-month salary management system, and store personal information (name, contact, address, etc.) of employees.
- Deliverables: Employee data storage, salary grading system, file handling, and user authentication.
- Tasks: System design, development in C++, testing, and documentation.
- Costs: Minimal, primarily development time.
- Deadlines: 1 week for planning, 3 weeks for development, 4–6 weeks for full completion.

List and briefly describe the scope of your project. Project scope is the part of project planning that involves determining and documenting a list of specific project goals, deliverables, tasks, costs and deadlines.]

1.4. Organization Of Project Report

[Organization of your full report. Write a simple clear brief summary of each chapter's contents and outcomes]

Chapter 2

BACKGROUND

2.1. Existing System Analysis

2.1.1:

Employee Management System(manual): In a manual system, employee information, such as personal details, attendance, and payroll, is recorded and managed by hand in physical records (files, spreadsheets, paper-based forms). This could involve using notebooks, spreadsheets, or printed documents.

pros:

Low Initial Cost : Requires no specialized equipment or programs.

Simple to Use: Managing records doesn't demand technical expertise.

Customization: Easily modified without coding or software skills..

cons:

prone to errors: Record loss is a possibility, and data entering errors can occur.

Time-consuming: It takes a while to generate reports or conduct information searches.

Lack of Scalability: Manual data management is harder and more complicated as the workforce size increases.

Analyzing data is difficult because without digital tools, reporting and analysis are less effective.

2.1.2:

School Management System (Digital) : A school management system (SMS) is a software solution designed to streamline various administrative tasks in educational institutions. Here are the pros and cons of implementing such a system:

pros:

Efficient administration: Automates administrative tasks, reduces paperwork and saves time.

Enhance communication: Promotes transparency in the education system by improving communication between teachers, students and parents.

cons:

Implementation Costs: Initial costs for software and training can be a significant investment for schools.

Resistance to Change: Staff or parents may resist adopting new technologies, impacting the system's effectiveness.

Technical Issues: Possible technical glitches or downtime can disrupt operations if not addressed promptly. [2]

Find some existing implemented /running system of your selected project. The system may be manual or digital. Study and briefly discuss some pros & cons of those systems.]

2.2. Research Gap:

1. User-Friendliness: Manual EMS and school management systems often lack intuitive interfaces. EMS should focus on simplicity and customization for different user roles to reduce training time.
2. Security and Data Privacy: Data security is sometimes overlooked by school management systems. To safeguard sensitive employee data, the Employee Management System must place a high priority on role-based access, robust encryption, and frequent audits.
3. Scalability: Scalability issues may arise with systems such as school management. EMS should be built to expand with the company, accommodating more staff members and positions.
4. Time-Consuming and Error-Prone: Manual data entry increases the chances of errors and slows down the booking process. EMS is a digitized system so it does not have these problems.

[Briefly write about the theoretical, mathematical, methodological, tools & technological knowledge you have applied in this project. You may write why you have chosen these knowledge, tools and techniques for your project development]

Chapter 3

SYSTEM ANALYSIS & DESIGN

3.1. Technology & Tools

[

The technology and tools utilized in the project are briefly described here:

Programming Language :

The project's primary programming language is C++. Because of its performance, flexibility, and support for object-oriented programming, C++ was selected.

Environment for Integrated Development (IDE) :

CodeBlocks: A well-known open-source integrated development environment for writing, compiling, and debugging C++ code. Multiple compilers are supported by Code::Blocks, which also offers an easy-to-use interface for productive development and debugging.

The compiler :

CodeBlocks' C++ code is compiled using the GNU GCC compiler. Many C++ features and optimizations are supported by this open-source compiler.

Hardware :

HP 250 G10 Notebook PC, 15.6-inch

This laptop, which has a dependable processor and a 15.6-inch display, offers a steady environment for C++ programming.

It has enough RAM and [SSD/HDD storage] to enable seamless testing, compilation, and coding.

Write a brief information about tools and technologies that used in this project development. Include Software/Hardware requirements, tools and other technologies & their versions brief in here.]

3.2. Model & Diagram

[Write model description and draw some logical design for your project. This part may contain following contents:]

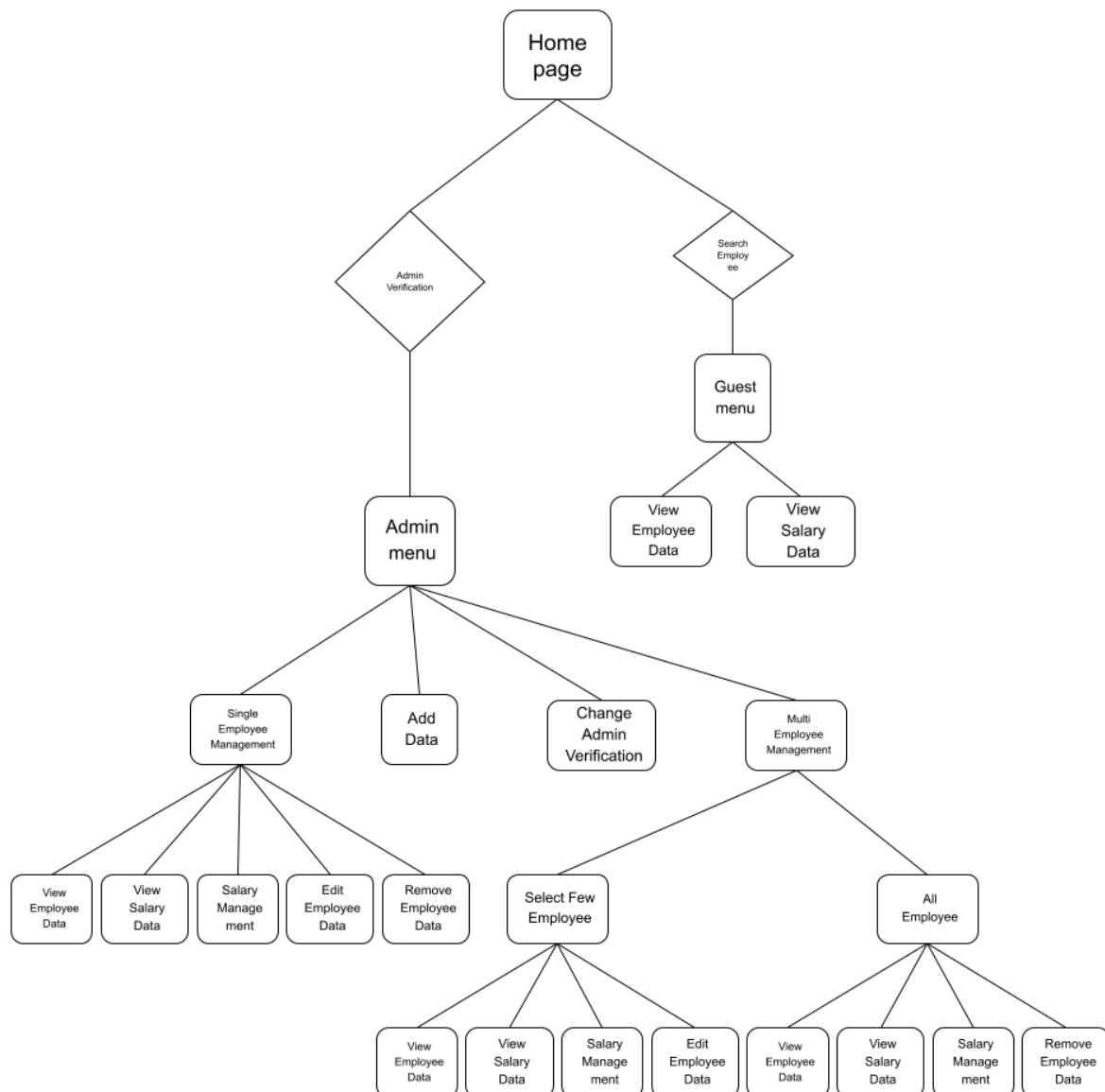


Figure 3.1: System Architecture Diagram

3.2.1. Model (SDLC/Agile/Waterfall/OOM)

[Discuss briefly any model that you have used to develop your entire project. Write its efficiency and benefits on your project.]

3.2.2. System Architecture

[If possible, draw a system architecture of your project. The system architecture diagram is an abstract depiction of the system's component architecture. It provides a succinct description of the system's component architecture in order to assist in component-component relationships and system functioning. Learn how to design system architecture from here:

<https://www.interviewbit.com/blog/system-architecture/>]

3.2.3. Use Case Diagram

[Draw a use case diagram for your project. You can study from the following link: <https://creately.com/guides/use-case-diagram-tutorial/> to know how to draw use case diagram]

3.2.4. Context Level Diagram

[Draw a context diagram that is known as DFD-0 of your project. Follow this link: <https://miro.com/blog/context-diagram/> to understand how to draw context diagram.]

3.2.5. Data Flow Diagram

[If your project contains data storage and flow of data from one to another process then design a Data Flow Diagram (DFD) a next version of context level diagram for your project. There can be more than one DFD level-1 of your project based on how many modules there are. Find a simple tutorial of drawing DFD-1 here: <https://www.lucidchart.com/pages/data-flow-diagram/how-to-make-a-dfd>]

3.2.6. Database Schema

[If your project contains database then provide a schema diagram. Learn how to draw schema from here: <https://www.educative.io/blog/what-are-database-schemas-examples>]

3.2.7. Algorithms/Flowchart

[Write any user defined algorithm or Draw flow chart of your proposed solution in the project. Take hints from this link: <https://www.lucidchart.com/pages/what-is-a-flowchart-tutorial>]

Chapter 4

IMPLEMENTATION

4.1. Interface/Front-End Design

[Discuss briefly how you designed the project interfaces, outlook, front-end part.]

4.2. Interface/Back-End Design

[Discuss briefly how you developed functionalities, back-end part, CRUD in your project.]

4.3. Modules/Features

[Lists the modules/Features you have developed in this project.]

Chapter 5

USER MANUAL

5.1. System Requirement

5.1.1. Hardware Requirement

Device - A Laptop
Operating System - Microsoft Windows 11 pro
RAM-16GB
Storage-512 GB

5.1.2. Software Requirement

The required software for the project are
IDE - Codeblocks
Compiler-GNU G++

5.2. User Interfaces

[Here is the detailed list and entitled of the project module with interface picture or screenshot.

List and entitled the project modules here with interface picture or screenshot. Also provide login or others credentials for user if needed.]

5.2.1. Panel A : Home page

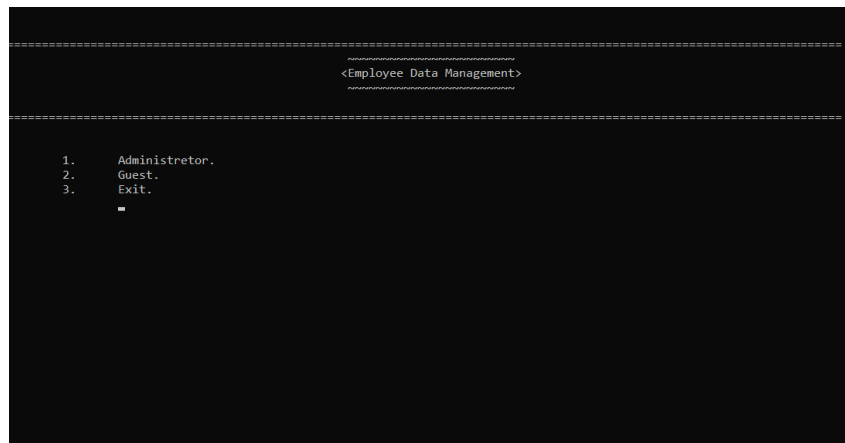


Figure 5.1

5.2.2. Panel B : Admin login interface (After pressing 1 in Panel A)

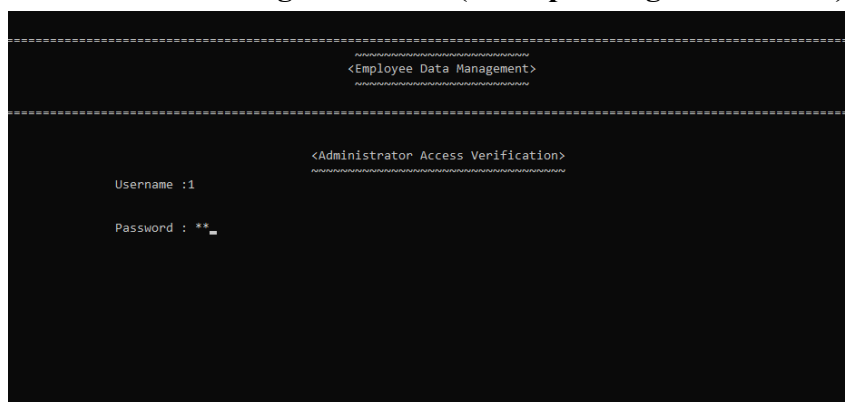


Figure 5.2

5.2.3. Panel C: Admin menu (After entering username and password in panel B)

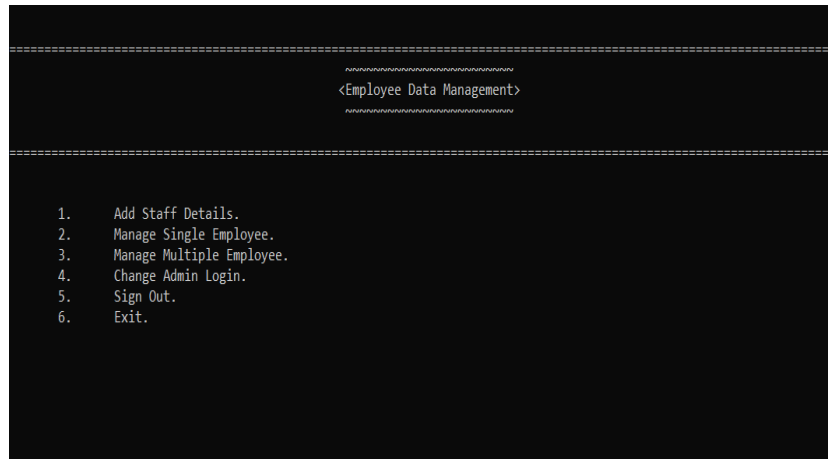


Figure 5.3

5.2.4. Panel D: Admin can add new staff details here(After pressing 1 in panel C)

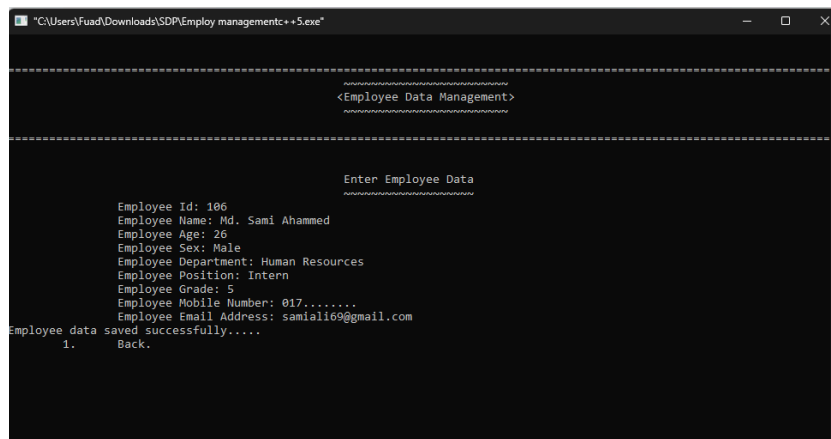


Figure 5.4

5.2.5. Panel E: Admin can search single employee details in this interface(After pressing 2 in panel C)

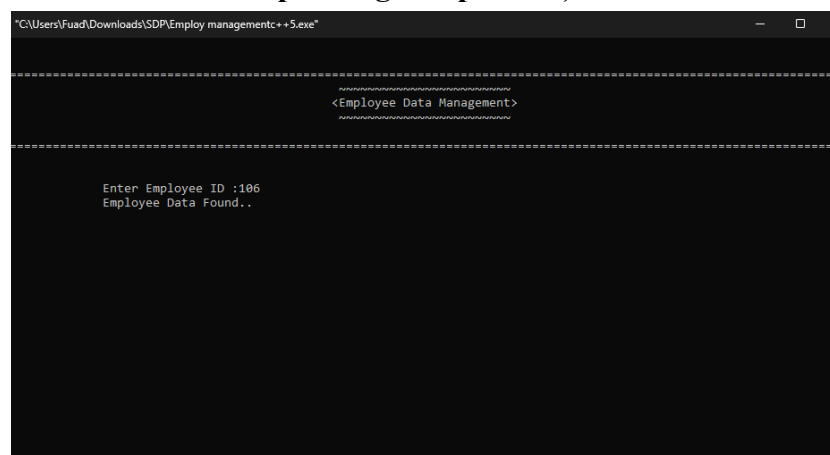


Figure 5.5

5.2.6. Panel F: Admin can handle single employee details in this interface(After entering employee id in panel E)

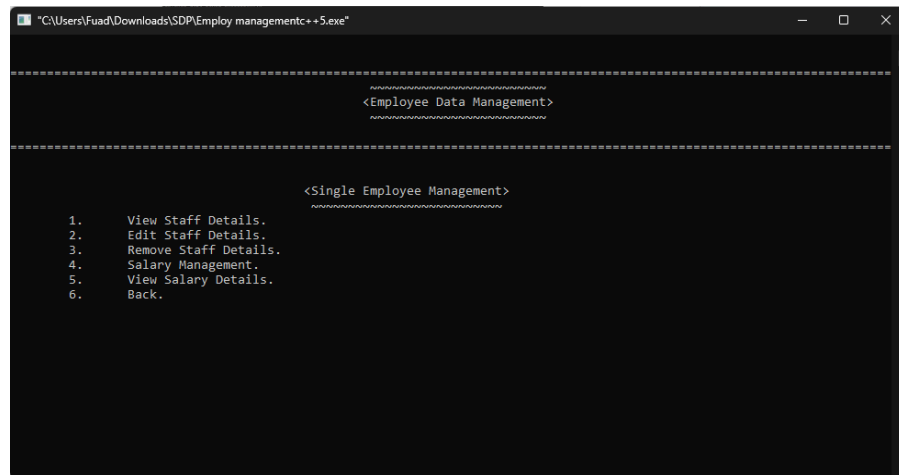


Figure 5.6

5.2.7. Panel G: Admin can view single employee details in this interface(After pressing 1 in panel F)

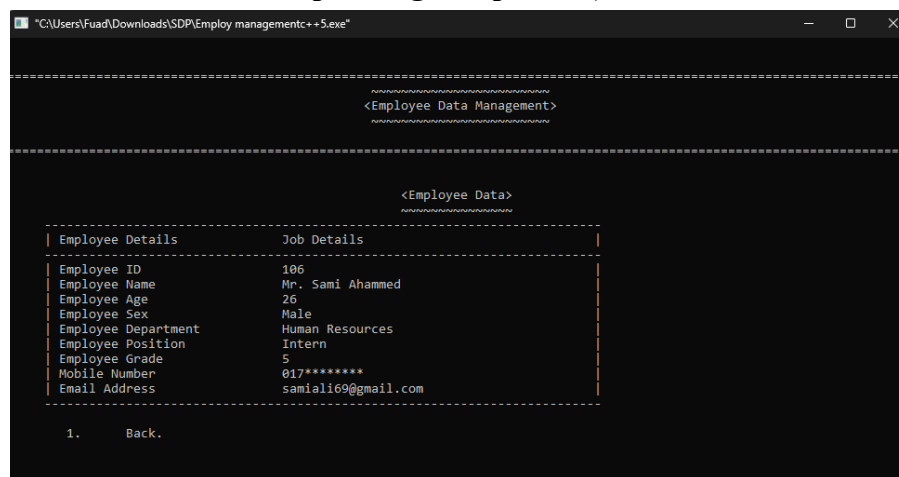


Figure 5.7

5.2.8. Panel H: Admin can edit single employee details in this interface(After pressing 2 in panel F)

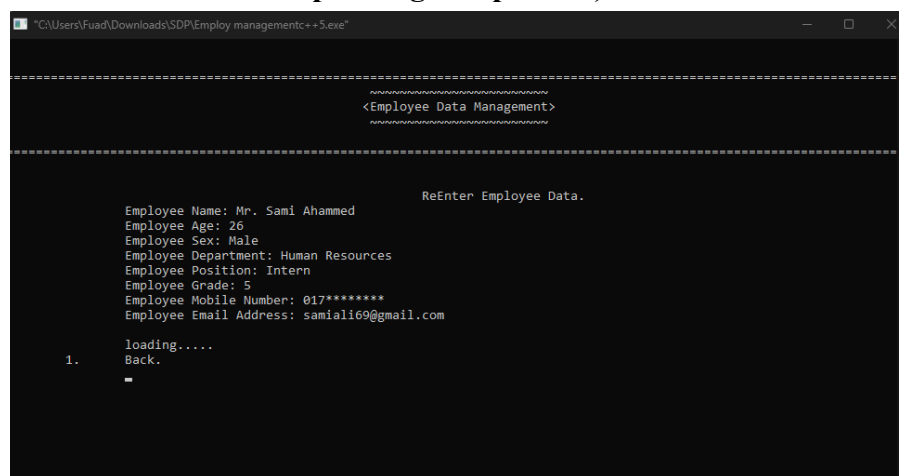


Figure 5.8

5.2.9. Panel I: Admin can remove single employee details in this interface(After pressing 3 in panel F)

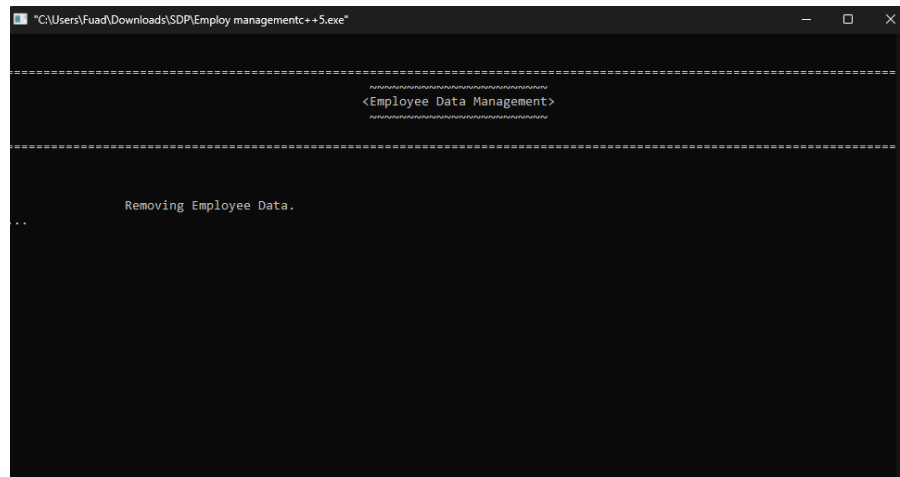


Figure 5.9

5.2.10. Panel J: Admin can handle single employee salary details in this interface (After pressing 4 in panel F)

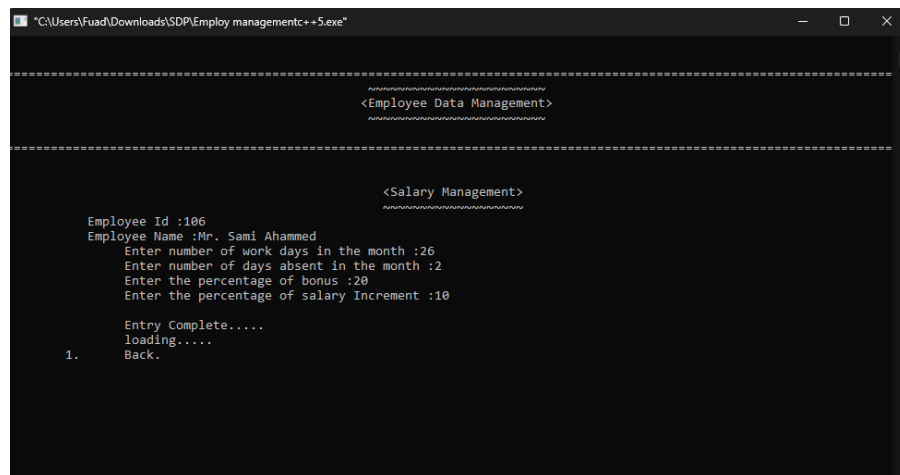


Figure 5.10

5.2.11. Panel K: Admin can view single employee salary details in this interface (After pressing 5 in panel F)

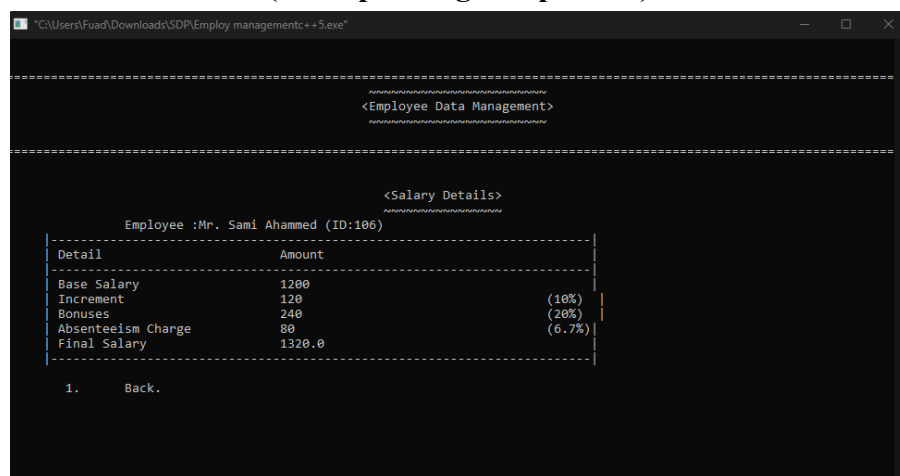


Figure 5.11

5.2.12. Panel L: Admin can search select few or all employees details in this interface(After pressing 3 in panel C)

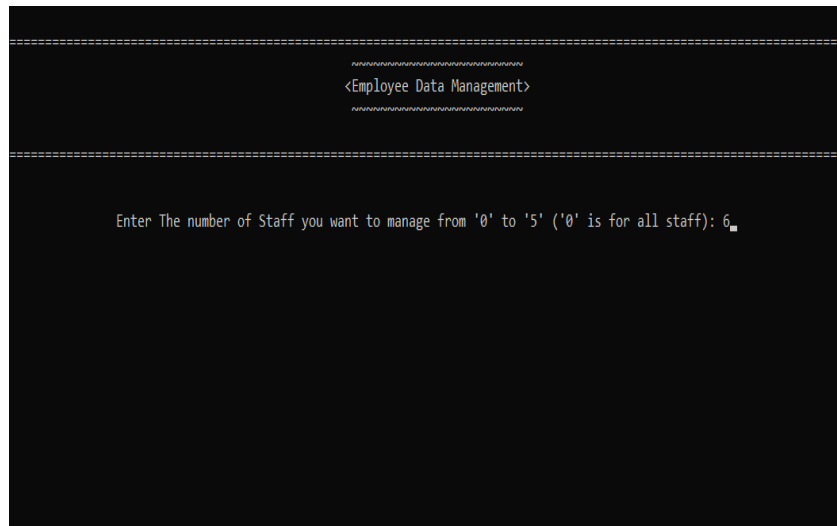


Figure 5.12

**5.2.13. Panel M: Admin can handle select few employees details in this interface
(After entering a number less than 5 in panel L)**

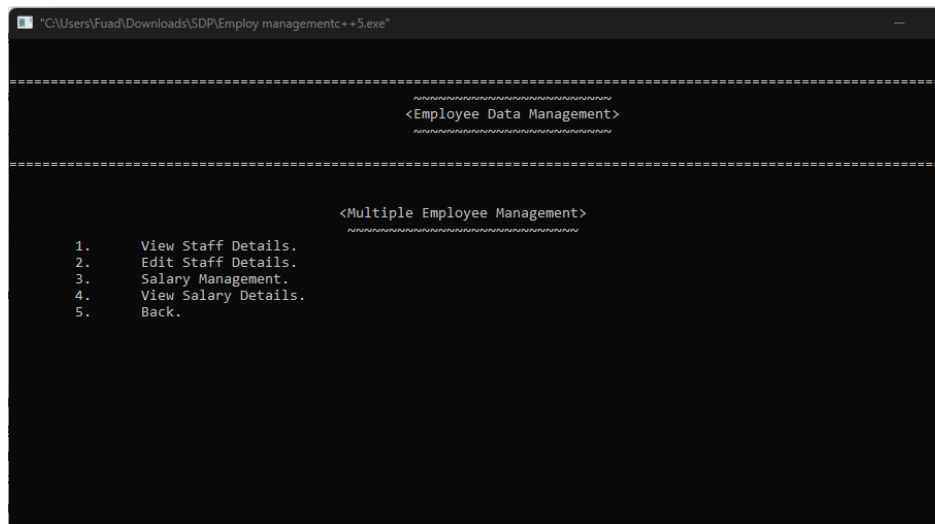


Figure :5.13

**5.2.14. Panel N: Admin can view select few employees details in this interface
(After pressing 1 in panel M)**

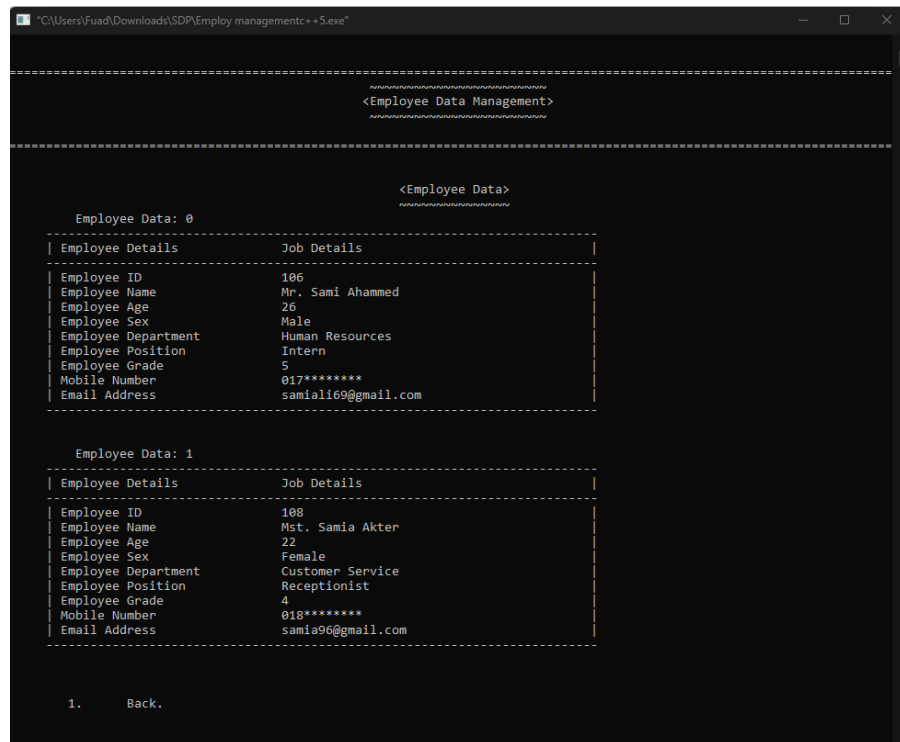


Figure 5.14

5.2.15. Panel O: Admin can edit select few employees details(After pressing 2 in panel M)

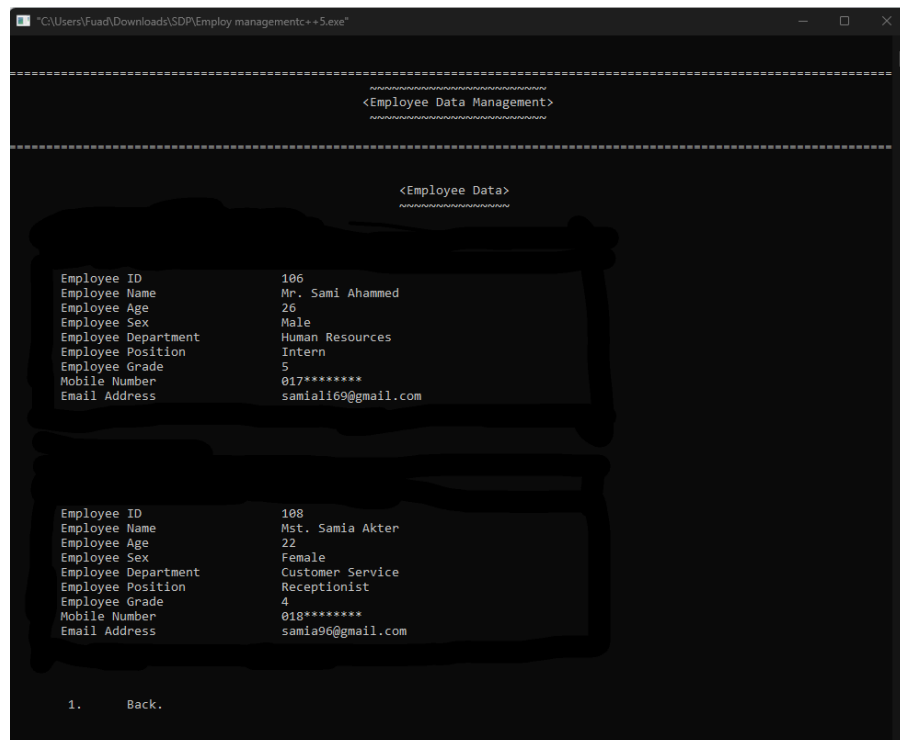


Figure 5.15

5.2.16. Panel P: Admin can handle select few employees salary details(After pressing 3 in panel M)

```

Users\Fuad\Downloads\SDP\Employ managementc++5.exe

=====
<Employee Data Management>
=====

<Salary Management>
=====

Employee Id :106
Employee Name :Mr. Sami Ahammed
Enter number of work days in the month :26
Enter number of days absent in the month :2
Enter the percentage of bonus :20
Enter the percentage of salary Increment :10

Entry Complete.....

Total Salary :1320

Employee Id :108
Employee Name :Mst. Samia Akter
Enter number of work days in the month :26
Enter number of days absent in the month :4
Enter the percentage of bonus :20
Enter the percentage of salary Increment :10

Entry Complete.....

Total Salary :1860

loading.....

Total Salary :3180

1. Back.

```

Figure 5.16

5.2.17. Panel Q: Admin can view select few employees salary details(After pressing 4 in panel M)

```

G:\Users\Fuad\Downloads\SDP\Employ managementc++5.exe

=====
<Employee Data Management>
=====

<Salary details>
=====

Employee :Mr. Sami Ahammed (ID:106)
-----
Detail                Amount
-----
Base Salary           1200
Increment              120                (10%)
Bonuses               240                (20%)
Absenteeism Charge     80                (6.7%)
Final Salary          1320.0

Employee :Mst. Samia Akter (ID:108)
-----
Detail                Amount
-----
Base Salary           1800.0
Increment              180.0                (10.0%)
Bonuses               360.0                (20.0%)
Absenteeism Charge     240.0                (13.3%)
Final Salary          1860.0

Total Salary :3180.0

1. Back.

```

Figure 5.17

5.2.18. Panel R: Admin can handle all employees details(After entering 0 in panel L)

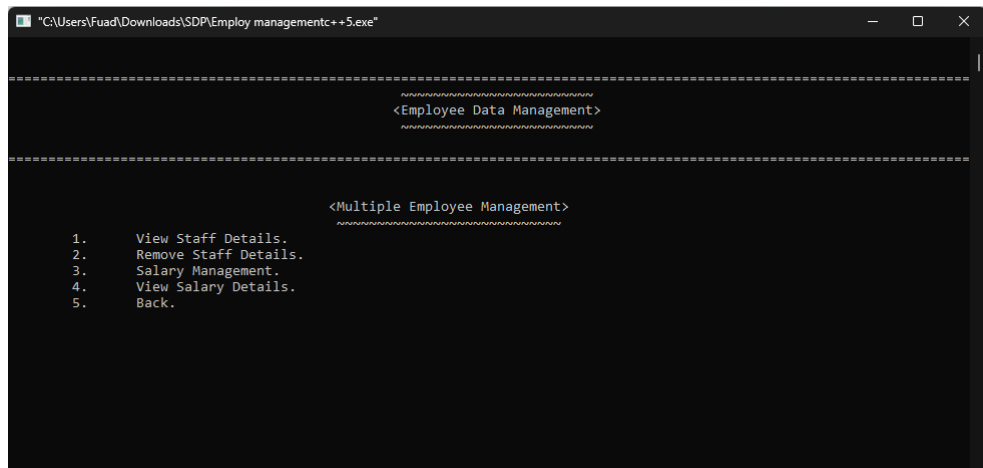


Figure 5.18

5.2.19. Panel S: Admin can change the username and password from this interface(After pressing 4 in panel C)

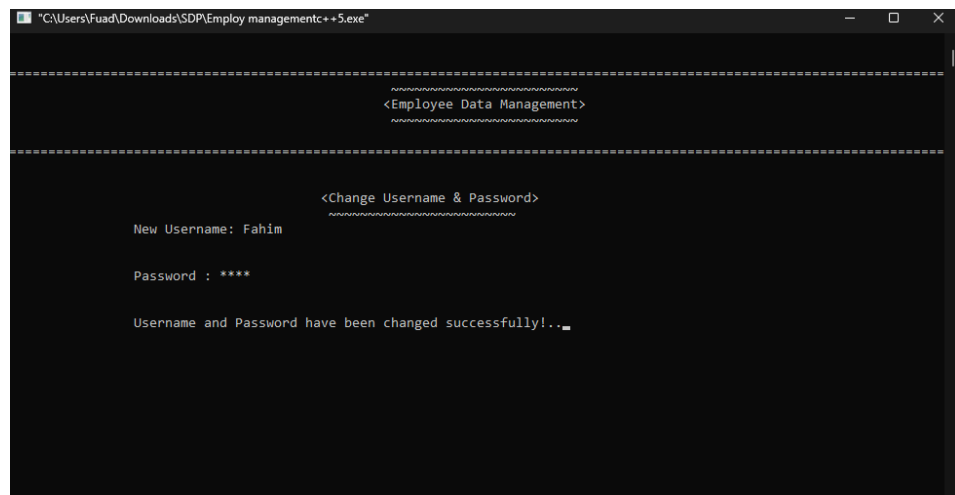


Figure 5.19

5.2.20. Panel T: Guest login interface(press 2 in panel A).

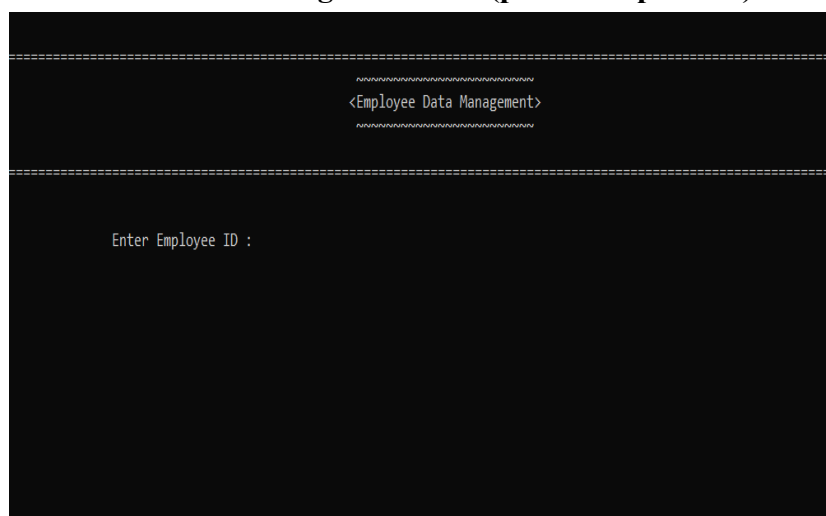


Figure 5.20

5.2.21. Panel U: Guest menu : Guests get two options where they can see their personal information and salary details.(After entering their ID in panel

T)

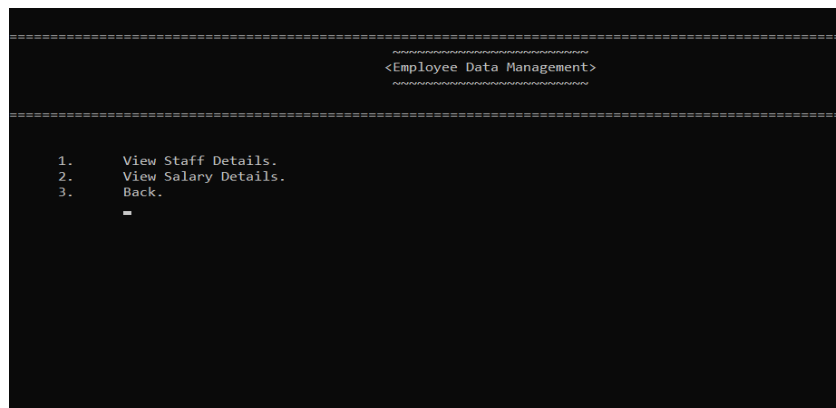


Figure 5.21

5.2.22. Panel V: Guest can see their personal information(after pressing 1 in Panel U)

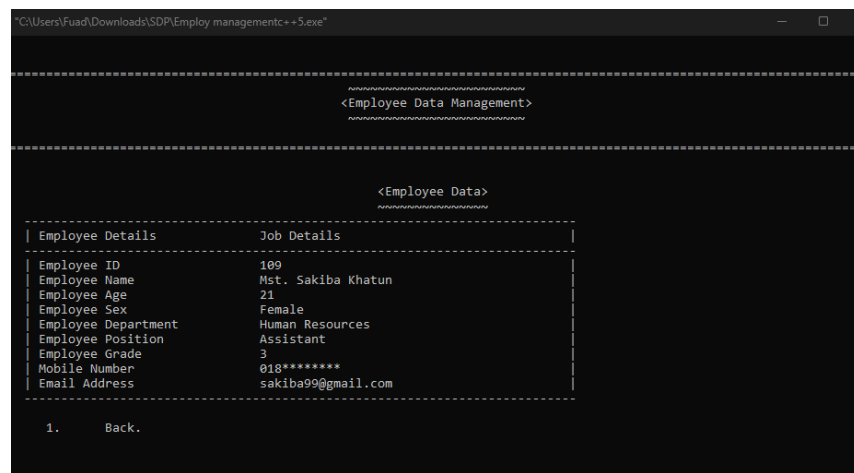


Figure 5.22

5.2.23. Panel W: Guest an option to view their monthly salary(after pressing 2 in panel U)

"C:\Users\Fuad\Downloads\SDP\Employee managementc++5.exe"

<Employee Data Management>

<Salary Details>

Employee :Mst. Sakiba Khatun (ID:109)

Detail	Amount	
Base Salary	2400	
Increment	0	(0%)
Bonuses	0	(0%)
Absenteeism Charge	0	(0.0%)
Final Salary	0.0	

1. Back.

Figure 5.23

Chapter 6

CONCLUSION

6.1. Conclusion

"Employee Management System" project has successfully created an easy-to-use and secure platform for managing employee information. The system helps improve work efficiency and supports the growth of the company with its simple features and smooth processes. The project not only addresses the current needs of the company but also sets the stage for future improvements.

6.2. Limitations

Here are some limitations of "Employee Management System" -

- Limited Number of Employees : The program struggles to handle a large number of employees and crashes when too much data is entered.
- Slow Performance : When there are too many employees record, the system takes more time to load or search data.
- Limited Features : The platform lacks advanced functionalities such as compensation processing, or employee evaluation monitoring, focusing instead on fundamental operations.
- No Backup System : If the system crashes there is no way to recover lost data, leading to potential data loss.
- Failure to Compute Annual Salary : The system is unable to calculate and record an employee's total yearly earnings (for all 12 months), which is a crucial feature for comprehensive payroll administration.

6.3. Future Works

As we finish the first phase of the "Employee Management System" project, there are many opportunities to improve and add new features to make the system even better for users. Here are some ideas for future updates:

- Multi- Platform Accessibility.
- Database Integration Setup.
- Use Advanced Software.
- Upgrade UI and UX .

References

[Write your references from where you have taken your definition, diagram, pictures, code or other contents. Use MLA format, you can use following link: <https://www.scribbr.com/citation/generator/> to generate automatically your citation.]

1. "What Is a Data Flow Diagram." *Lucidchart*, www.lucidchart.com/pages/data-flow-diagram.
2. TA Consult. (n.d.). *What is bank management?* Retrieved December 8, 2024, from <https://www.taconsult.biz/en/knowledge/what-is-bank-management/>

Appendix A